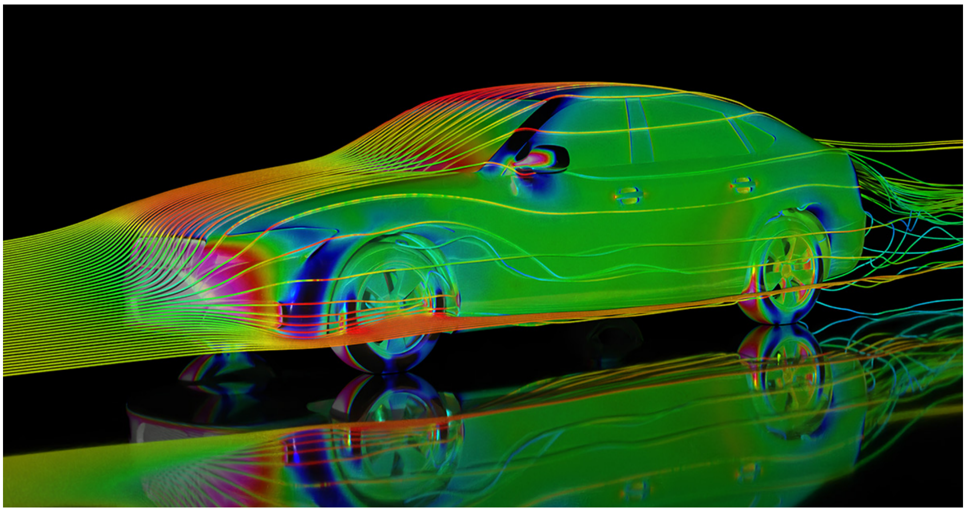

LMECA2660 - Numerical Methods in Fluid Mechanics

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https://github.com/SimonDesmidt/Syntheses

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UCLouvain

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Introduction - Finite differences with uniform grid

1.1 Classical finite differences

Let us define a function $u(\cdot)$ that depends on a variable x. Suppose that in the dimension x, we discretize the function uniformly with a step h and the values at the nodes are written u_i. Then, by a Taylor development series,

$$\begin{cases} u_{i+1} = u_i + h\left(\frac{\partial u}{\partial x}\right)_i + \frac{h^2}{2!}\left(\frac{\partial^2 u}{\partial x^2}\right)_i + \frac{h^3}{3!}\left(\frac{\partial^3 u}{\partial x^3}\right)_i + \frac{h^4}{4!}\left(\frac{\partial^4 u}{\partial x^4}\right)_i + \dots \\ u_{i-1} = u_i - h\left(\frac{\partial u}{\partial x}\right)_i + \frac{h^2}{2!}\left(\frac{\partial^2 u}{\partial x^2}\right)_i - \frac{h^3}{3!}\left(\frac{\partial^3 u}{\partial x^3}\right)_i + \frac{h^4}{4!}\left(\frac{\partial^4 u}{\partial x^4}\right)_i - \dots \end{cases} \tag{1.1}$$

This gives three possible finite-difference approximations:

$$\begin{aligned} \left(\frac{\partial u}{\partial x}\right)_i &= \frac{u_{i+1} - u_i}{h} + \mathcal{O}(h) && \text{(Forward differences)} \\ \left(\frac{\partial u}{\partial x}\right)_i &= \frac{u_i - u_{i-1}}{h} + \mathcal{O}(h) && \text{(Backward differences)} \\ \left(\frac{\partial u}{\partial x}\right)_i &= \frac{u_{i+1} - u_{i-1}}{2h} + \mathcal{O}(h^2) && \text{(Centered differences)} \end{aligned} \tag{1.2}$$

This also gives, for the second order,

$$\left(\frac{\partial^2 u}{\partial x^2}\right)_i = \frac{u_{i+1} - 2u_i + u_{i-1}}{h^2} - \frac{h^2}{12}\left(\frac{\partial^4 u}{\partial x^4}\right)_i + \dots \tag{1.3}$$

$\rightarrow$ Note: in general, discentered differences are only used for stability reasons.

1.2 Richardson extrapolation

Richardson extrapolation combines centered finite differences at different scales to get a better error:

$$\begin{aligned} \frac{4}{3}\Bigg[\left(\frac{\partial u}{\partial x}\right)_i &= \frac{u_{i+1} - u_{i-1}}{2h} - \frac{h^2}{6}\left(\frac{\partial^3 u}{\partial x^3}\right)_i - \frac{h^4}{120}\left(\frac{\partial^5 u}{\partial x^5}\right)_i - \dots \Bigg] \\ \frac{-1}{3}\Bigg[\left(\frac{\partial u}{\partial x}\right)_i &= \frac{u_{i+2} - u_{i-2}}{2(2h)} - \frac{(2h)^2}{6}\left(\frac{\partial^3 u}{\partial x^3}\right)_i - \frac{(2h)^4}{120}\left(\frac{\partial^5 u}{\partial x^5}\right)_i - \dots \Bigg] \\ \implies \left(\frac{\partial u}{\partial x}\right)_i &= \frac{8(u_{i+1} - u_{i-1}) - (u_{i+2} - u_{i-2})}{12h} + \frac{h^4}{30}\left(\frac{\partial^5 u}{\partial x^5}\right)_i - \dots \end{aligned} \tag{1.4}$$

With this method, the truncation error is of order $\mathcal{O}(h^4)$. In the same way, for second order,

$$\left(\frac{\partial^2 u}{\partial x^2}\right)_i = \frac{4}{3}\frac{u_{i+1} - 2u_i + u_{i-1}}{h^2} - \frac{1}{3}\frac{u_{i+2} - 2u_i + u_{i-2}}{(2h)^2} + \mathcal{O}(h^4) \tag{1.5}$$

1.3 Operators

Let us define the following operators:

- Forward difference: $\Delta u_i = u_{i+1} - u_i$;
- Backward difference: $\nabla u_i = u_i - u_{i-1}$;
- Centered difference: $\delta u_i = u_{i+1/2} - u_{i-1/2}$;
- Mean: $\mu u_i = \frac{1}{2}(u_{i+1/2} + u_{i-1/2})$;

→ Note: $u_{i+1/2}$ and $u_{i-1/2}$ are not computable because they are not grid values, but can be used for derivations of other formulae.

- Identity operator: $I u_i = u_i$;
- Forward operator: $E u_i = u_{i+1}$;
- Backward operator: $E^{-1} u_i = u_{i-1}$;

→ Note: $E^{-1}E = I$.

Those operators have the following properties:

- $\mu\delta = \frac{1}{2}(E - E^{-1})$;
- $\mu^2 = I + \delta^2/4$;

The forward operator can be re-expressed using a Taylor development series:

$$\begin{aligned} E u_i = u_{i+1} &= u_i + h\frac{\partial}{\partial x}u_i + \frac{h^2}{2!}\frac{\partial^2}{\partial x^2}u_i + \frac{h^3}{3!}\frac{\partial^3}{\partial x^3}{}_i + \dots \\ &= \left(I + hD + \frac{(hD)^2}{2!} + \frac{(hD)^3}{3!} + \dots\right) u_i = \exp(hD)u_i \end{aligned} \tag{1.6}$$

From this, using a second Taylor development series,

$$hD = \log(I + \Delta) = \Delta - \frac{\Delta^2}{2} + \frac{\Delta^3}{3} - \frac{\Delta^4}{4} + \dots \tag{1.7}$$

And, in the same way,

$$hD = -\log(I - \nabla) = \nabla + \frac{\nabla^2}{2} + \frac{\nabla^3}{3} + \frac{\nabla^4}{4} + \dots \tag{1.8}$$

We can do the same for another operator:

$$\mu\delta = \frac{1}{2}(E - E^{-1}) = \frac{1}{2}\left(\exp(hD) - \exp(-hD)\right) = \sinh(hD) \tag{1.9}$$

and we can use the Taylor series for $arc\,\sinh(x)$ but it is not very useful. By the property that $\mu^2 = I + \delta^2/4$, we get another form:

$$hD = \mu\delta\left(I - \frac{1}{6}\delta^2 + \frac{1}{30}\delta^4 - \frac{140}{\delta^6} + \dots\right) \tag{1.10}$$

If we keep only the first order term, we find the centered-difference scheme, and the terms up to second order give the Richardson extrapolation.

→ Note: in any scheme, using more information (more values, e.g. $u_{i+2}, u_{i+3}, \dots$) gives a more accurate solution and the order of the truncation error increases (e.g. to $\mathcal{O}(h^3)$).

1.4 2D Laplacian

For finite differences in 2D, we can define several types of stencils. For a second-order error, there is the cross operator, which is simply the sum of classical centered finite differences on both axes, and the box operator. This operator is a linear combination of the cross operator using the medians of the square, and the one that uses the diagonals of the square (see 1.1).

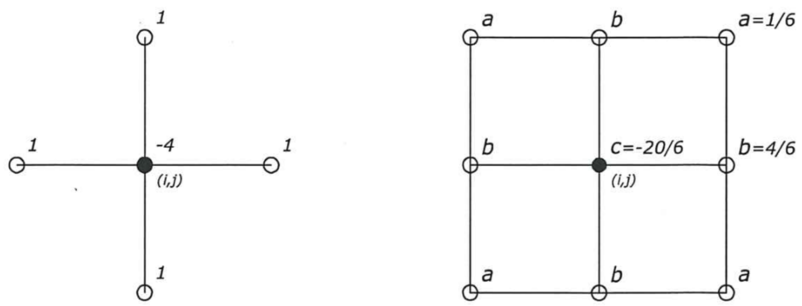


Figure 1.1: Cross operator and box operator.

The box operator expresses the following quantity:

$$h^2\nabla^2\left(u + \frac{h^2}{12}\nabla^2 u + \dots\right) = h^2\left(\nabla^2 u + \frac{h^2}{12}\nabla^2(\nabla^2 u)\right) \tag{1.11}$$

Those stencils can be generalized to higher orders using more points (bigger cross and bigger square).

→ Note: the coefficients are found using the constraint that the truncation error is independent of orientation of the stencil.

Space integration

2.1 Convection equation

The convection equation is

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0 \tag{2.1}$$

The analytic solution of this equation on an infinite domain, for a speed c constant, is

$$u(x,t) = A(t)e^{ikx} = A(0)e^{ik(x-ct)} \tag{2.2}$$

On a periodic domain of length L, the solution is

$$\begin{aligned} u(x,t) = \sum_{k=-\infty}^{\infty} A_k(t)e^{ikx} \qquad k = \frac{2\pi}{L}p \qquad p \in \mathbb{Z} \\ \Longrightarrow u(x,t) = \sum_{p=-\infty}^{\infty} A_p(t)e^{i\frac{2\pi x}{L}} \end{aligned} \tag{2.3}$$

where the coefficients verify the condition $A_k = A^*_{-k}$ for all k, since the solution must be real. In the exact solution, all the modes have the same speed c.

2.1.1 Explicit schemes

However, all modes do not have the same velocity when we use explicit finite differences. Let us show it in the case of an infinite domain (same thing happens for a periodic domain, adding the sum on p):
Let $u_i(t) = A(t)e^{jkx_i}$. Then,

$$\begin{aligned} \left.\frac{\partial u}{\partial x}\right|_i = Ajk^* \exp(jkx_i) \Longrightarrow \frac{dA}{dt} + jk^*cA = 0 \\ \Longrightarrow u_i(t) = A(0)e^{j(kx_i - k^*ct)} = A(0)e^{ik\left(x_i - \frac{k^*h}{kh}ct\right)} \end{aligned} \tag{2.4}$$

we call k^* the modified wave number. It is different from k because all the modes do not move at the same speed. For example, for the E2 stencil, its expression is derived in the following way:

$$\begin{aligned} \left.\frac{\partial u}{\partial x}\right|_i = \frac{u_{i+1} - u_{i-1}}{2h} = \frac{A}{h}\frac{1}{2}(e^{jkh} - e^{-jkh})e^{jkx_i} = \frac{A}{h}j\sin(kh)e^{ikx_i} = Ajk^*e^{jkx_i} \\ \Longrightarrow k^*h = \sin(kh) \end{aligned} \tag{2.5}$$

This is a **phase** error, as the modes do not **move** with the right **velocity**.
By a Taylor development,

$$\frac{k^*h}{kh} = 1 - \frac{(kh)^2}{6} + \mathcal{O}((kh)^4) \tag{2.6}$$

and although k is not constant for all modes, the error of the stencil is still of the same order: $\mathcal{O}((kh)^2)$. For higher order explicit scheme (E4, E6,...), the error is of the same order as the scheme. Moreover, the speed of the modes is $c^* = \frac{k^*h}{kh}c$.

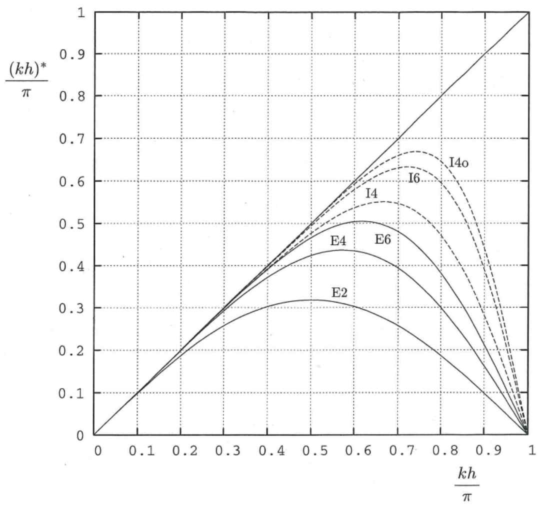


Figure 2.1: Evolution of the exact wave number and the modified wave number.

As the error increases when k increases, it is important to use a very refined grid so that all points whose amplitude is non negligible have $k^*h \approx kh$.

2.1.2 Implicit schemes

The general scheme of implicit finite differences is the following:

$$\begin{aligned} \frac{\partial u}{\partial x}_i + \alpha\frac{1}{2}\left(\left.\frac{\partial u}{\partial x}\right|_{i+1} + \left.\frac{\partial u}{\partial x}\right|_{i-1}\right) + \beta\frac{1}{2}\left(\left.\frac{\partial u}{\partial x}\right|_{i+2} + \left.\frac{\partial u}{\partial x}\right|_{i-2}\right) \\ = a\frac{u_{i+1} - u_{i-1}}{2h} + b\frac{u_{i+2} + u_{i-2}}{4h} + c\frac{u_{i+3} - u_{i-3}}{6h} \end{aligned} \tag{2.7}$$

Usually, we use $\beta = c = 0$ to keep only the nearest grid points to i. Those schemes are called compact.
We can use $\left.\frac{\partial u}{\partial x}\right|_i = Ajk^*e^{jkx_i}$ and $\frac{u_{i+1}-u_{i-1}}{2h} = \sin(kh)$ to show that

$$k^*h = \frac{a\sin(kh) + \frac{b}{2}\sin(2kh) + \frac{c}{3}\sin(3kh)}{1 + \alpha\cos(kh) + \beta\cos(2kh)} \tag{2.8}$$

We can do a Taylor development of this quantity to get different schemes:

$$\begin{aligned} \text{Taylor of order 1:} \quad & 1 + \alpha + \beta = a + b + c \Longrightarrow \text{Error of order 2} \\ \text{Taylor of order 2:} \quad & 3(\alpha + 2^2\beta) = a + 2^2b + 3^2c \Longrightarrow \text{Error of order 4} \\ \text{Taylor of order 3:} \quad & 5(\alpha + 2^4\beta) = a + 2^4b + 3^4c \Longrightarrow \text{Error of order 6} \\ \text{Taylor of order 4:} \quad & 7(\alpha + 2^6\beta) = a + 2^6b + 3^6c \Longrightarrow \text{Error of order 8} \\ \text{Taylor of order 5:} \quad & 9(\alpha + 2^8\beta) = a + 2^8b + 3^8c \Longrightarrow \text{Error of order 10} \end{aligned} \tag{2.9}$$

For example, in the case where $\beta = 0$ and $\alpha \neq 0$, the system is tridiagonal and the solver has a time complexity of $\mathcal{O}(N)$. For $\beta = 0$ and $\alpha = 0$, we find the explicit scheme.

→ Note: The I4o scheme is the scheme with the smallest constant before the $(kh)^4$ using all parameters. For this one, the sign of that constant is positive, meaning that it goes above the correct velocity. This behaviour is not present on the other scheme.

2.1.3 One-sided finite differences

Let us consider the backward differences for the convection equation:

$$\frac{d}{dt}u_i = -c\frac{u_i - u_{i-1}}{h} \Longrightarrow \frac{d}{dt}A_k = -\frac{c}{h}(1 - e^{jkh}) \tag{2.10}$$

Then, $\lambda_k = -\frac{c}{h}(1 - e^{-jkh})$ is no longer purely imaginary. Its real part gives an amplitude error (numerical diffusion) and its imaginary part gives a (numerical) dispersion error.

→ Note: for $c < 0$, we need to use forward differences instead because otherwise the real part of λ_k becomes positive and thus unstable.

Let us compute the term of numerical diffusion:

$$\frac{d}{dt}u_i = -c\frac{u_i - u_{i-1}}{h} = -c\frac{u_{i+1} - u_{i-1}}{2h} + c\frac{u_{i+1} - 2u_i + u_{i-1}}{2h} = -c\frac{u_{i+1} - u_{i-1}}{2h} + \frac{ch}{2}\frac{u_{i+1} - 2u_i + u_{i-1}}{h^2} \tag{2.11}$$

The modified equation becomes

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = \frac{ch}{2}\frac{\partial^2 u}{\partial x^2} + \dots \tag{2.12}$$

and so the diffusivity parameter is thus $\epsilon = \frac{ch}{2}$.
For this scheme, from Equation (2.10), $\lambda_k \Delta t = -\frac{c\Delta t}{h}\left((1 - \cos(kh)) + i\sin(kh)\right)$ and the region of stability of the method is a circle centered in $(-1, 0)$ in the complex plane of $\lambda_k \Delta t$.

2.1.4 One-sided finite differences of higher order

This type of scheme of higher order would give for example

$$\begin{aligned} \frac{d}{dt}u_i &= \frac{3u_i - 4u_{i-1} + u_{i-2}}{2h} \\ &= \frac{11u_i - 18u_{i-1} + 9u_{i-2} - 2u_{i-3}}{6h} \end{aligned} \tag{2.13}$$

and all those expressions give different stability regions. The general form for the spatial derivative is

$$\left.\frac{\partial u}{\partial x}\right|_i = a_1\frac{u_i - u_{i-1}}{h} + a_2\frac{u_i - u_{i-2}}{2h} + a_3\frac{u_i - u_{i-3}}{3h} \tag{2.14}$$

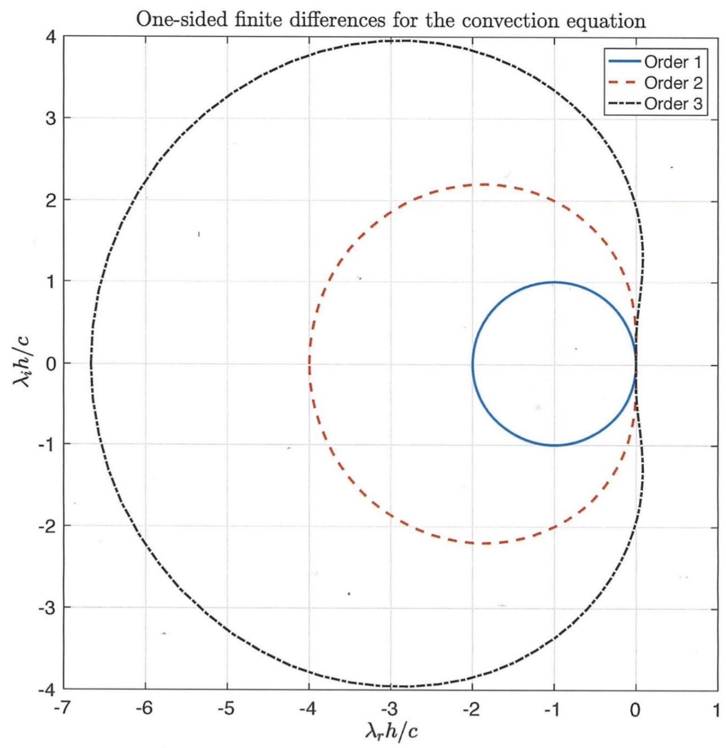


Figure 2.2: Stability regions for one-sided schemes.

This means that if our system has a high eigenvalue λ, we need to decrease the time step to have convergence properties. This means that those one-sided schemes are not much helpful, because they are very costly. Moreover, at order 3, we can see low wave numbers with positive real part, meaning that they blow up (positive exponent in the exponential).

2.2 Diffusion equation

The diffusion equation is

$$\frac{\partial u}{\partial t} = \alpha \frac{\partial^2 u}{\partial x^2} \tag{2.15}$$

This equation is not reversible because it represents a loss of information: the high-wave-number terms decay fast and have no more impact after a small time.
The exact solution is $u(x,t) = A_k(t)e^{jkx}$ with $A_k(t) = A_k(0)e^{\alpha k^2 t}$.

2.2.1 Explicit schemes

The explicit finite differences scheme is

$$\left.\frac{\partial^2 u}{\partial x^2}\right|_i = a\frac{u_{i+1} - 2u_i + u_{i-1}}{h^2} + b\frac{u_{i+2} - 2u_i + u_{i-2}}{4h^2} + c\frac{u_{i+3} - 2u_i + u_{i-3}}{9h^2} \tag{2.16}$$

Let $u_i(t) = A(t)e^{jkx_i}$. Then,

$$\frac{dA}{dt} = -\alpha(k^2)^* A \Longrightarrow A(t) =< A(0)e^{-\alpha(k^2)^* t} e^{jkx_i} \Longrightarrow u_i(t) = A(0)e^{-\alpha k^2 \left(\frac{\left(k^2\right)^* h^2}{k^2 h^2}\right)} e^{jkx_i} \tag{2.17}$$

This is a **amplitude** error, as the mode do not decay with the proper **rate**.
For example, for the explicit scheme of order 2 ($a = 1$, $b = c = 0$), the modified k^2 is given by

$$(k^2)^* h^2 = 4\sin^2\left(\frac{kh}{2}\right) \Longleftrightarrow \frac{(k^2)^* h^2}{k^2 h^2} = \frac{\sin^2\left(\frac{kh}{2}\right)}{\left(\frac{kh}{2}\right)^2} \tag{2.18}$$

2.2.2 Implicit schemes

The general form of the implicit scheme follows the same reasoning as for Equation (2.7).

$$\begin{aligned} &\left.\frac{\partial^2 u}{\partial x^2}\right|_i + \alpha\frac{1}{2}\left(\left.\frac{\partial^2 u}{\partial x^2}\right|_{i+1} + \left.\frac{\partial^2 u}{\partial x^2}\right|_{i-1}\right) + \beta\frac{1}{2}\left(\left.\frac{\partial^2 u}{\partial x^2}\right|_{i+2} + \left.\frac{\partial^2 u}{\partial x^2}\right|_{i-2}\right) \\ &= a\frac{u_{i+1} - 2u_i + u_{i-1}}{h^2} + b\frac{u_{i+2} - 2u_i + u_{i-2}}{4h^2} + c\frac{u_{i+3} - 2u_i + u_{i-3}}{9h^2} \end{aligned} \tag{2.19}$$

By using $\left.\frac{\partial^2 u}{\partial x^2}\right|_i = -A(t)(k^2)^* e^{jkx_i}$, we can show that

$$(k^2)^* h^2 = \frac{4\left[a\sin^2\left(\frac{kh}{2}\right) + \frac{b}{4}\sin^2\left(\frac{2kh}{2}\right) + \frac{c}{9}\sin^2\left(\frac{3kh}{2}\right)\right]}{1 + \alpha\cos(kh) + \beta\cos(2kh)} \tag{2.20}$$

As previously, we can do the Taylor development series of this expression to get different schemes:

$$\begin{aligned} \text{Taylor of order 1:} \quad & 1 + \alpha + \beta = a + b + c \Longrightarrow \text{Error of order 2} \\ \text{Taylor of order 2:} \quad & 2\cdot 3(\alpha + 2^2\beta) = a + 2^2 b + 2^2 c \Longrightarrow \text{Error of order 4} \\ \text{Taylor of order 3:} \quad & 3\cdot 5(\alpha + 2^4\beta) = a + 2^4 b + 2^4 c \Longrightarrow \text{Error of order 6} \\ \text{Taylor of order 4:} \quad & 4\cdot 7(\alpha + 2^6\beta) = a + 2^6 b + 2^6 c \Longrightarrow \text{Error of order 8} \\ \text{Taylor of order 5:} \quad & 5\cdot 9(\alpha + 2^8\beta) = a + 2^8 b + 2^8 c \Longrightarrow \text{Error of order 10} \end{aligned} \tag{2.21}$$

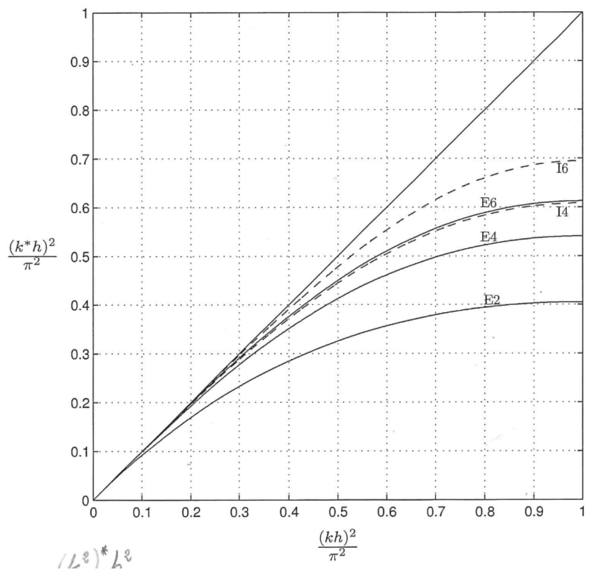


Figure 2.3: Evolution of the modified k^2 with k^2.

2.3 Convection-diffusion equation

The convection-diffusion equation is

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = \epsilon \frac{\partial^2 u}{\partial x^2} \tag{2.22}$$

→ Note: numerical diffusion can appear and pollute the physical diffusion. This is not a problem if we have $\epsilon_{num} \ll \epsilon_{phys}$.

2.3.1 Centered finite differences

The scheme here is

$$\frac{d}{dt}u_i = -c\frac{u_{i+1} - u_{i-1}}{2h} + \epsilon\frac{u_{i+1} - 2u_i + u_{i-1}}{h^2} \tag{2.23}$$

This gives

$$\lambda_k = -\frac{2\epsilon}{h^2}(1 - \cos(kh)) - j\frac{c}{h}\sin(kh) \tag{2.24}$$

This expression in the complex plane of $\lambda_k \Delta t$ is an ellipse with vertical axis $b = \beta$ and horizontal axis $a = 2r$, $\beta = c\Delta t/h$ being the CFL number, and $r = \epsilon\Delta t/h^2$ being the Fourier number. We also define the mesh Peclet number $R = |c|h/\epsilon$, giving the identity $\beta = rR$.
This scheme, using a Euler scheme for integration, is stable if the ellipse is fully contained in the circle centered in $(-1, 0)$.

- If $b \leq a$, the scheme is stable if $a \leq 1$, or equivalently, if $R \leq 2$, the scheme is stable if $r \leq 1/2$;

- If $b > a$, the scheme is stable if the largest radius of curvature of the ellipse is ≤ 1[1], i.e. $b^2/a \leq 1 \iff c^2\Delta t/\epsilon \leq 2$, or equivalently, if $R > 2$, this scheme is stable for $r \leq 2/R^2$.

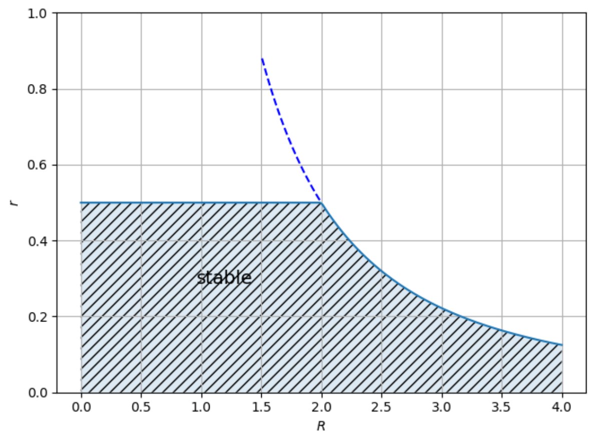


Figure 2.4: Stability region for the parameters of the convection-diffusion equation.

If using backward finite differences, i.e. $c > 0$, the scheme is

$$\frac{d}{dt}u_i = -c\frac{u_i - u_{i-1}}{h} + \epsilon\frac{u_{i+1} - 2u_i + u_{i-1}}{h^2} = -c\frac{u_{i+1} - u_{i-1}}{2h} + \left(\epsilon + \frac{ch}{2}\right)\frac{u_{i+1} - 2u_i + u_{i-1}}{h^2} \tag{2.25}$$

Then,

$$\lambda_k = -\left(\frac{c}{h} + \frac{2\epsilon}{h^2}\right)(1 - \cos(kh)) - j\frac{c}{h}\sin(kh) \tag{2.26}$$

This is also an ellipse, with axes $a = \beta + 2r$ and $b = \beta$. Here, $b \leq a$ becaue $r > 0$ always, and thus

$$r \leq \frac{1}{1+R} \tag{2.27}$$

[1]This value is the radius of curvature of the circle.

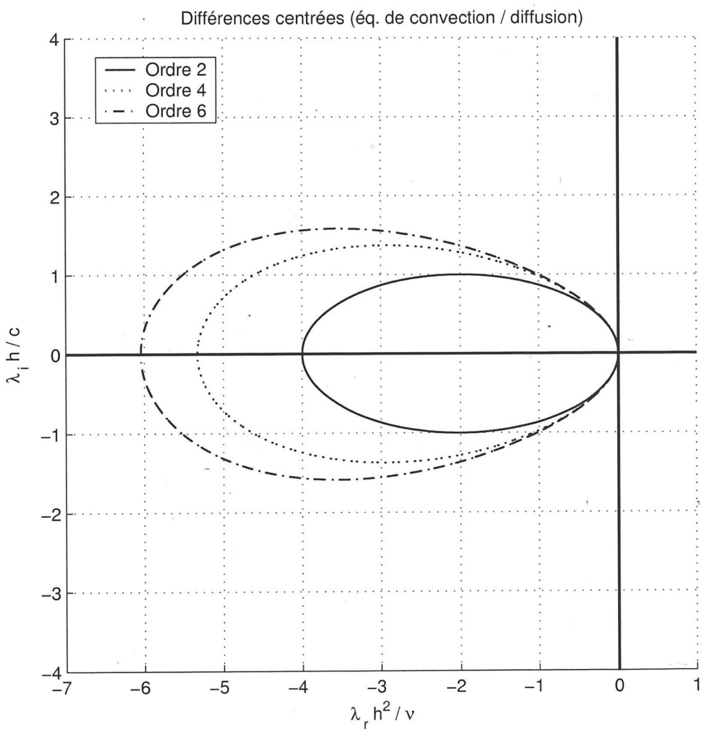


Figure 2.5: Stability region of the convection-diffusion equation.

Diffusion with non constant ϵ. The handling of this non constant term is always done in the same manner. For example, for the E2 scheme,

$$\begin{aligned}\frac{\partial}{\partial x}\left(\epsilon\frac{\partial u}{\partial x}\right)\bigg|_i &= \frac{\epsilon_{i+1/2}(u_{i+1}-u_i)-\epsilon_{i-1/2}(u_i-u_{i-1})}{h^2} \\ \epsilon_{i+1/2} &= \frac{1}{2}\left(\epsilon(x_i,u_i)+\epsilon(x_{i+1},u_{i+1})\right)\end{aligned} \tag{2.28}$$

2.4 Convection equation in 2D

The convection equation in 2D has two different velocity values:

$$\frac{\partial u}{\partial t}+u_0\frac{\partial u}{\partial x}+v_0\frac{\partial u}{\partial y}=0 \tag{2.29}$$

We consider here a periodic problem to be able to do a modal analysis. The E2 scheme gives

$$\frac{du_{ij}(t)}{dt}=-\left(u_0\frac{u_{i+1,j}-u_{i-1,j}}{2\Delta x}+v_0\frac{u_{i,j+1}-u_{i,j-1}}{2\Delta y}\right) \tag{2.30}$$

Which gives a system of ODEs for the $u_{i,j}(t)$.
In the continuous problem, the solution is

$$u(x,y,t)=\sum_{k,\ell}A_{k\ell}(t)e^{j(kx+\ell y)} \tag{2.31}$$

Our ODE gives

$$A_{k\ell}(t)=A_{k\ell}(0)e^{-j(u_0k+v_0\ell)t} \tag{2.32}$$

For the discrete problem (in space but continuous time), the ODE for $A_{k\ell}$ is

$$\frac{dA_{k\ell}}{dt} = -j\left(\frac{u_0}{\Delta x}\sin(k\Delta x) + \frac{v_0}{\Delta y}\sin(l\Delta y)\right)A_{k\ell} \tag{2.33}$$

and so the eigenvalues $\lambda_{k\ell}$ are purely imaginary, with $\lambda_{k\ell} = \lambda_k + \lambda_\ell$. We can also define the global CFL number

$$\beta = \beta_x + \beta_y = \frac{|u_0|\Delta t}{\Delta x} + \frac{|v_0|\Delta t}{\Delta y} \tag{2.34}$$

and

$$|\lambda_{k\ell}|_{\max}\Delta t = \beta \tag{2.35}$$

2.5 Convection-diffusion equation in 2D

The convection-diffusion equation in 2D is handled in the same way it is in 1D but using two parts. There is no conceptual difference.

$$\frac{\partial u}{\partial t} + u_0\frac{\partial u}{\partial x} + v_0\frac{\partial u}{\partial y} = \epsilon\left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2}\right) \tag{2.36}$$

We can compute the eigenvalues by using a scheme, e.g. E2, to get

$$\frac{dA_{k\ell}}{dt} = \lambda_{k\ell}A_{k\ell} \tag{2.37}$$

The parameters of the scheme are

$$\begin{cases} r_x = \frac{\epsilon\Delta t}{(\Delta x)^2} & R_x = \frac{|u_0|\Delta x}{\epsilon} \Longrightarrow \beta_x = r_x R_x \\ r_y = \frac{\epsilon\Delta t}{(\Delta y)^2} & R_y = \frac{|v_0|\Delta y}{\epsilon} \Longrightarrow \beta_y = r_y R_y \end{cases} \tag{2.38}$$

In the case $\Delta x = \Delta y = h$, $r_x = r_y = r = \epsilon\Delta t/h^2$ and

$$\begin{cases} \beta_x = \frac{|u_0|\Delta t}{h} \\ \beta_Y = \frac{|v_0|\Delta t}{h} \end{cases} \Longrightarrow \begin{cases} \beta = \beta_x + \beta_y = \frac{(|u_0|+|v_0|)\Delta t}{h} \\ R = R_x + R_y = \frac{(|u_0|+|v_0|)h}{\epsilon} \end{cases} \tag{2.39}$$

Numerical time-integration schemes for ODEs

The general formulation of first order ODEs is

$$\frac{du}{dt} = f(u,t) \tag{3.1}$$

Until now, we have only had a simple system with $f(u,t) = \lambda u$.
We can do a Taylor development of $f(\cdot)$ to analyze the local behaviour of the system:

$$\frac{du}{dt} = f(u,t) = \underbrace{f(u_0,t_0)}_{=:\beta} + (u-u_0)\underbrace{\frac{\partial f}{\partial u}}_{=:\lambda} + (t-t_0)\underbrace{\frac{\partial f}{\partial t}(u_0,t_0)}_{=:\gamma} + \dots \tag{3.2}$$

and defining $u' = u - u_0$ and $t' = t - t_0$, we get

$$\frac{du'}{dt'} - \lambda u' = \beta + \gamma t'. \tag{3.3}$$

The solution of this is, with initial condition $u'(t'=0) = u(0) - u_0 = 0$,

$$u'(t) = u'_p + u'_g = -\frac{1}{\lambda}\left(\gamma t' + \left(\frac{\gamma}{\lambda} + \beta\right)\right) + Ae^{\lambda t'} = \frac{1}{\lambda}\left[\left(\frac{\gamma}{\lambda} + \beta\right)\left(e^{\lambda t'} - 1\right) - \gamma t'\right] \tag{3.4}$$

The stability of the scheme used to integrate that ODE is governed by the homogeneous part of the equation, as it is the only one that can blow up (exponential term).

For our special case $\frac{du}{dt} = \lambda u$, the solution is $u(t) = u(0)e^{\lambda t}$. As we use a discretized space, let $t = t^n = n\Delta t$. Then, $u^n = u(t^n) = (e^{\lambda \Delta t})^n u^0 = (\rho_e)^n u^0$, where we call ρ_e the amplification factor of the exact solution. The ODE is (physically) stable if $|\rho_e| \leq 1$. In the $\lambda \Delta t$ complex plane, it represents the left half-plane.

3.1 Explicit Euler scheme

$$\frac{u^{n+1} - u^n}{\Delta t} = \lambda u^n \tag{3.5}$$

Then,

$$u^{n+1} = (1 + \lambda \Delta t)u^n = \rho u^n \Longrightarrow u^n = (\rho)^n u^0 \tag{3.6}$$

The scheme is stable if
$$|\rho|^2 = (1+\lambda_r \Delta t)^2 + (\lambda_i \Delta t)^2 \leq 1 \tag{3.7}$$
This is a circle centered in $(-1, 0)$.

At each time stpe, the Euler scheme produces an $\mathcal{O}((\Delta t)^2)$ error. To arrive at $t = T$, it takes $N = T/\Delta t$ time steps. The error at $t = T$ is thus $\mathcal{O}((\Delta t)^2/\Delta t) = \mathcal{O}(\Delta t)$. The Euler scheme is a first-order scheme.

→ Note: always check the order of a scheme using a fixed time interval, and not a single time step, as we want a global estimation.

If λ is large in modulus but in the stable part of the physical solution, then decreasing the time step can make $\lambda \Delta t$ lie in the stability region.

3.2 Implicit Euler scheme

$$\frac{u^{n+1} - u^n}{\Delta t} = \lambda u^{n+1} \tag{3.8}$$
Then,
$$u^{n+1} = \frac{1}{1 - \lambda \Delta t} u^n = \rho u^n \tag{3.9}$$
This scheme is unconditionnally stable, as it is stable for all problems with $\lambda_r \leq 0$ and we do not consider the problems that are not physically stable.

3.3 Trapezoid rule

$$\frac{u^{n+1} - u^n}{\Delta t} = \lambda \frac{u^{n+1} - u^n}{2} \tag{3.10}$$
Then,
$$u^{n+1} = \frac{1 + \frac{\lambda \Delta t}{2}}{1 - \frac{\lambda \Delta t}{2}} u^n = \rho u^n \tag{3.11}$$
This scheme is also unconditionnally stable for physically stable problems. Moreover, it is of second order, because it is centered around $t^{n+1/2}$.

3.4 Explicit Runge-Kutta schemes

We will work here on the integration of ODEs of the form
$$\frac{du}{dt} = f(u, t) \tag{3.12}$$
The notation is $f^n = f(u^n, t^n)$ for the value of the function at the n-th time step.
The general form of a Runge-Kutta scheme of order m is the following:
$$u^{n+1} = u^n + \sum_{i=1}^{m} \gamma_i k_i \qquad \text{s.t.} \qquad \begin{cases} k_1 = \Delta t f(u^n, t^n) \\ k_i = \Delta t f(u^n + \sum_{j=1}^{i-1} \beta_{ji} k_j, t^n + \sum_{j=1}^{i-1} \alpha_j \Delta t) & \forall i \geq 2 \end{cases} \tag{3.13}$$

3.4.1 RK2

The formulation of order two is

$$u^{n+1} = u^n + \gamma_1 k_1 + \gamma_2 k_2 \qquad \text{s.t.} \qquad \begin{cases} k_1 = \Delta t f(u^n, t^n) \\ k_2 = \Delta t f(u^n + \beta k_1, t^n + \alpha \Delta t) \end{cases} \tag{3.14}$$

The explicit Runge-Kutta schemes work with Taylor series:

$$u^{n+1} = u^n + \Delta t \left(\frac{du}{dt}\right)^n + \frac{(\Delta t)^2}{2}\left(\frac{d^2u}{dt^2}\right)^n + \mathcal{O}((\Delta t)^3) \tag{3.15}$$

and

$$\begin{aligned} k_2 &= \Delta t \left[f^n + \beta k_1 \left(\frac{\partial f}{\partial u}\right)^n + \alpha \Delta t \left(\frac{\partial f}{\partial t}\right)^n + \dots \right] \\ \implies u^{n+1} &= u^n + \gamma_1 \Delta t f^n + \gamma_2 \Delta t \left[f^n + \beta \Delta t f^n \left(\frac{\partial f}{\partial u}\right)^n + \alpha \Delta t \left(\frac{\partial f}{\partial t}\right)^n + \dots \right] \end{aligned} \tag{3.16}$$

By comparing those developments, we find a system of 3 equations:

$$\begin{aligned} \gamma_1 + \gamma_2 &= 1 \\ \gamma_2 \beta &= 1/2 \\ \gamma_2 \alpha &= 1/2 \end{aligned} \tag{3.17}$$

There remains one degree of freedom, meaning we get a family of schemes. The usual methods are

- RK2 classic: Heun's method

$$\begin{aligned} \gamma_1 = \gamma_2 &= 1/2 \\ \alpha = \beta &= 1 \end{aligned} \tag{3.18}$$

$$\begin{cases} k_1 = \Delta t f(u^n, t^n) \\ k_2 = \Delta t f(u^n + k_1, t^n + \Delta t) \end{cases} \implies u^{n+1} = u^n + \frac{1}{2}k_1 + \frac{1}{2}k_2 \tag{3.19}$$

- RK2 classic: midoint method

$$\begin{aligned} \gamma_1 = 0 \quad \gamma_2 &= 1 \\ \alpha = \beta &= 1/2 \end{aligned} \tag{3.20}$$

$$\begin{cases} k_1 = \Delta t f(u^n, t^n) \\ k_2 = \Delta t f(u^n + \frac{1}{2}k_1, t^n + \frac{1}{2}\Delta t) \end{cases} \implies u^{n+1} = u^n + k_2 \tag{3.21}$$

3.4.2 RK3-4

Runge-Kutta 3 and 4 are to be derived in the same way. They both have two degrees of freedom and thus it is possible to have many different approaches to get the coefficients. The most common way is to do a diagonal scheme, i.e. k_j only depends on the value of k_{j-1}. This gives the usual RK4 method

$$\begin{cases} k_1 = \Delta t f(u^n, t^n) \\ k_2 = \Delta t f(u^n + \frac{1}{2}k_1, t^n + \frac{1}{2}\Delta t) \\ k_3 = \Delta t f(u^n + \frac{1}{2}k_2, t^n + \frac{1}{2}\Delta t) \\ k_4 = \Delta t f(u^n + k_3, t^n + \Delta t) \end{cases}$$
$$u^{n+1} = u^n + \frac{1}{6}(k_1 + 2k_2 + 2k_3 + k_4) \tag{3.22}$$

3.4.3 Runge Kutta schemes in stability analysis

The analysis of stability is for the equation

$$\frac{du}{dt} = \lambda u \tag{3.23}$$

Stability only considers the linearized equation, because it analyses the local behaviour. In the case of this equation, we can express u^{n+1} as a function of u^n:

$$u^{n+1} = \rho u^n \tag{3.24}$$

where $\rho = \sum_{i=1}^{m} \frac{(\lambda \Delta t)^m}{m!}$ for a Runge-Kutta scheme of order m.
The method is stable if $|\rho| \leq 1$.

→ Note: the intersection of the marginal stability curve ($|\rho| = 1$) with the imaginary axis is particularly relevant, as it is necessary to determine the maximum value of the CFL for simple spatial-integration schemes such as Euler explicit, because the λ values of those schemes are purely imaginary. In short, the stability region of the spatial scheme must be inside the stability region of the stability region of the time scheme to be fully stable.

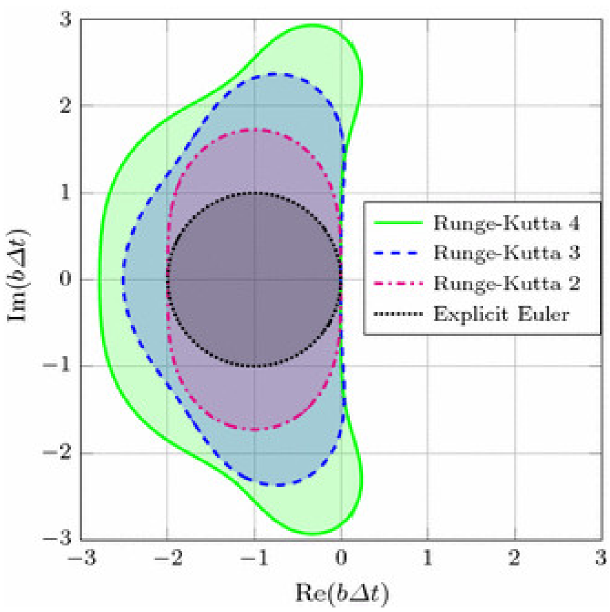


Figure 3.1: Stability regions of Runge-Kutta schemes.

3.4.4 General Explicit Runge-Kutta schemes

A Butcher tableau is a representation of the RK coefficients meant for them to be easy to read.

$$\begin{array}{c|ccc|c} & & & & \gamma_1 \\ \alpha_2 & \beta_{21} & & & \gamma_2 \\ \alpha_3 & \beta_{31} & \beta_{32} & & \gamma_3 \\ \alpha_4 & \beta_{41} & \beta_{42} & \beta_{43} & \gamma_4 \end{array}$$

The first column contains the time weights, the β are the space weights, and the γ are the weights in the linear combination of k_i.
We can check that we always have $\alpha_i = \sum_j \beta_{ji}$ and that $\sum_i \gamma_i = 1$.

→ Note: the diagonal schemes mentioned earlier give a diagonal matrix for the parameters β.

In an Explicit Runge-Kutta scheme (ERK), the number of substeps (number of coefficients k) is not necessarily the order of the scheme: it is possible to do ERK54, which has 5 substeps, but only an order 4. This is because the expression of the amplification paramater ρ changes, and does not correspond exactly to the Taylor development terms.

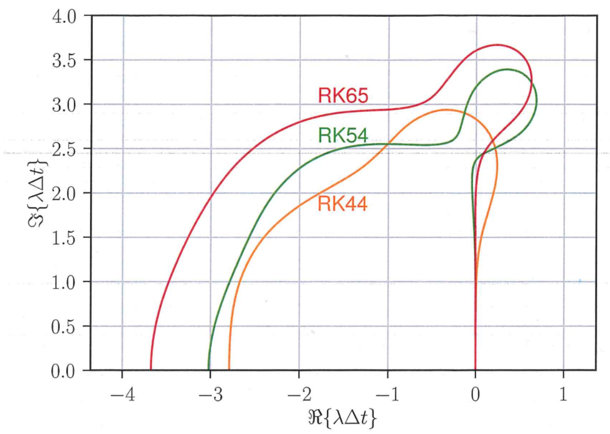


Figure 3.2: Stability regions for Explicit Runge-Kutta schemes.

You can see on 3.2 that the scheme RK54 goes to the right on the imaginary axis. This is a bad behaviour for stability, as there is a small physically stable zone that is not numerically stable.

3.4.5 RK4 pseudocode

Algorithm 1 RK4 pseudocode

(t, u) = time t^n and solution $u(t^n) = u^n$;
Save to avoid overwriting: $u_{safe} = u$, $t_{safe} = t$;
for $k = 1, 2, 3, 4$ **do**
  $u_\ell = u_{safe} + \alpha_k \Delta u$
  $t_\ell = t_{safe} + \alpha_k \Delta t$
  Evaluate: $\Delta u = \Delta t f(u_\ell, t_\ell)$
  And update: $\begin{cases} u = u + \gamma_k \Delta u \\ t = t + \gamma_k \Delta t \end{cases}$
end for
return (t, u)

At the end of the loop, (t, u) are the time $t^{n+1} = t^n + \Delta t$ and the solution $u(t^{n+1}) = u^{n+1}$. This algorithm needs 4 arrays to be stored. For big simulations, this is a lot of data, and this is why there exists some low-storage RK methods that only require 3 arrays. They use different paramater values, e.g. for RK3:

			5/30
1/3	1/3		9/30
3/4	-3/16	15/16	16/30

3.5 Multistep time schemes

Multisteps time schemes use more than one previous step to compute the value t^n. They are generally cheap, but not self starting: t^1 cannot be computed through it, as it would need to know t^{-1}. Moreover, they produce multiple solutions even though only one is physically correct.

3.5.1 Leap-frog

The scheme is

$$\frac{u^{n+1} - u^{n-1}}{2\Delta t} = f(u^n, t^n) \iff u^{n+1} = u^{n-1} + 2\Delta t f(u^n, t^n) \tag{3.25}$$

The scheme is centered around t^n, meaning it is of order 2.
To analyse its stability region, we have

$$\frac{du}{dt} = \lambda u \implies \begin{cases} u^n = \rho u^{n-1} \\ u^{n+1} = \rho^2 u^{n-1} \end{cases} \tag{3.26}$$

Then,

$$\rho^2 = 1 + 2\lambda \Delta t \rho \iff \begin{cases} \rho_1 + \rho_2 = 2\lambda \Delta t \\ \rho_1 \rho_2 = -1 \end{cases} \implies \rho = \lambda \Delta t \pm \sqrt{1 + (\lambda \Delta t)^2} \tag{3.27}$$

By Taylor series,

$$\begin{cases} \rho_1 = 1 + \lambda\Delta t + \frac{(\lambda\Delta t)^2}{2} - \frac{(\lambda\Delta t)^4}{8} + \cdots \approx e^{\lambda\Delta t} \\ \rho_2 = -1 + \lambda\Delta t - \frac{(\lambda\Delta t)^2}{2} + \frac{(\lambda\Delta t)^4}{8} + \cdots \approx e^{-\lambda\Delta t} \end{cases} \tag{3.28}$$

ρ_1 is the physical root and ρ_2 is called the parasitic root. Because of the condition $\rho_1\rho_2 = -1$ and the stability condition $|\rho_1|, |\rho_2| \leq 1$, we need $|\rho_1| = |\rho_2| = 1$.

Then, the solution of the scheme is

$$\rho = c_1\rho_1 + c_2\rho_2 \Longrightarrow u^n = (c_1\rho_1 + c_2\rho_2)u^{n-1} \tag{3.29}$$

where u^1 must be computed using another scheme with less steps, e.g. RK2.

3.5.2 General multi-step schemes

The general form of multi-step schemes is the following:

$$\sum_{m=0}^{k} \alpha_m u^{n+1-m} = \Delta t \sum_{m=0}^{k} \beta_m f(u^{n+1-m}, t^{n+1-m}) \tag{3.30}$$

If $\beta_0 = 0$, then the scheme is explicit, and implicit if not.

Adams-Bashforth (EXPLICIT)

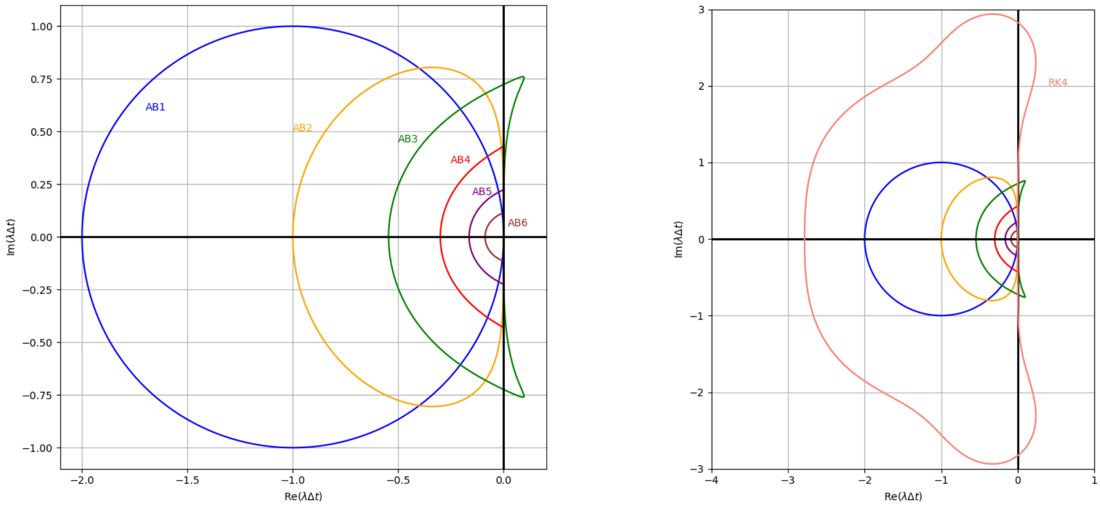
The Adams Bashforth schemes are a class of explicit multi-step schemes of the following form:

$$u^{n+1} = u^n + \Delta t \sum_{m=1}^{k} \beta_m f^{n+1-m} \tag{3.31}$$

The first three schemes are

- $k = 1$ (explicit Euler): $u^{n+1} = u^n + \Delta t f^n$;
- $k = 2$: $u^{n+1} = u^n + \frac{\Delta t}{2}(3f^n - f^{n-1})$;
- $k = 3$: $u^{n+1} = u^n + \frac{\Delta t}{12}(23f^n - 16f^{n-1} + 5f^{n-2})$.

When k increases, the stability factor ρ has more and more roots: the function $\Gamma(\rho)$ such that $u^{n+1} = \Gamma(\rho)u^n$ has k roots. The stability regions of the Adams-Bashforth methods are shown on Figure 3.3. Higher order means better accuracy, but it is also stable on a much smaller region, and all of them are far smaller than the stability region of Runge-Kutta methods. However, they require far less evaluation per step than RK methods.



(a) Adams-Bashforth from 1 to 6.

(b) Adams-Bashforth compared to RK4.

Figure 3.3: Stability regions of Adams-Bashforth methods

Adams-Moulton (IMPLICIT)

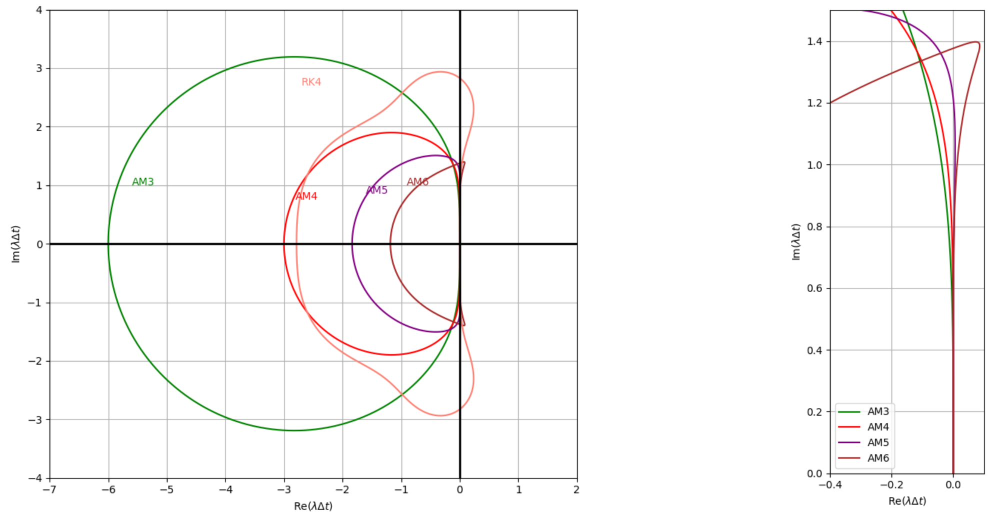
The Adams-Moulton schemes are a class of implicit multi-step schemes of the following form:

$$u^{n+1} = u^n + \Delta t \sum_{m=0}^{k-1} \beta_m f^{n+1-m} \tag{3.32}$$

The first three schemes are

- $k = 1$ (implicit Euler): $u^{n+1} = u^n + \Delta t f^{n+1}$;
- $k = 2$ (Crank Nicholson): $u^{n+1} = u^n + \frac{\Delta t}{2}(f^{n+1} + f^n)$;
- $k = 3$: $u^{n+1} = u^n + \frac{\Delta t}{12}(5f^{n+1} + 8f^n - f^{n-1})$.

The number of roots of the stability factor is $k - 1$ in these schemes, meaning that both $k = 1$ and $k = 2$ are unconditionnally stable. The stability regions are shown on Figure 3.4.



(a) Adams-Moulton from 3 to 6 (1 and 2 are unconditionnally stable) compared to RK4.

(b) Zoom of Adams-Moulton to show that 5 and 6 go to the right of the imaginary axis.

Figure 3.4: Stability regions of Adams-Bashforth methods.

Hybrid Adams methods

The idea of the hybrid methods is to use AB as a predictor and AM and a corrector, as AB is easy to use (explicit) but has poor stability, and AM has the opposite.
Using AB1 as predictor and AM2 as corrector yields Runge-Kutta 2. Other example using AB4 and AM4:

$$\begin{aligned} u^{*,n+1} &= u^n + \frac{\Delta t}{24}\left(55f^n - 59f^{n-1} + 37f^{n-2} - 9f^{n-3}\right) \\ u^{n+1} &= u^n + \frac{\Delta t}{24}\left(9f^{*,n+1} + 19f^n - 5f^{n-1} + f^{n-2}\right) \end{aligned} \tag{3.33}$$

This 4$^{\text{th}}$ order scheme requires only two evaluations of f, compared to 4 for RK4 of the same error order.

→ Note: the predictor can be done with one order less than the corrector without changing the order of the scheme.

A family of multi-steps second order explicit schemes

$$u^{n+1} = \alpha u^n + (1-\alpha)u^{n-1} + \frac{\Delta t}{2}\left((4-\alpha)f^n - \alpha f^{n-1}\right) \tag{3.34}$$

The case $\alpha = 0$ returns the leapfrog scheme (see Equation (3.25)) and the case $\alpha = 1$ returns AB2. The scheme is order 2 for any value of α except $\alpha = -4$: it is order 3 but unstable.

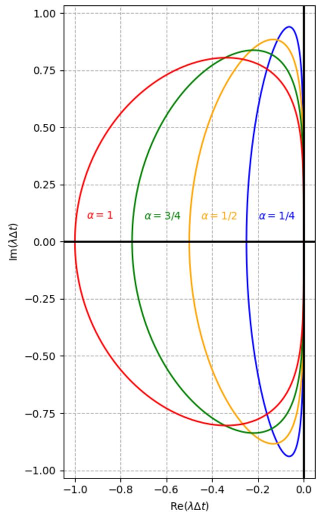


Figure 3.5: Stability regions of the family of multi-steps second order explicit schemes.

3.5.3 3 point backward differentiation formula

The 3BDF scheme does a Taylor series development of u^n and u^{n-1} around u^{n+1}. It yields

$$\frac{3u^{n+1} - 4u^n + u^{n-1}}{2\Delta t} = \left(\frac{du}{dt}\right)^{n+1} - \frac{1}{3}(\Delta t)^2 \left(\frac{d^3u}{dt^3}\right)^{n+1} + \cdots = f^{n+1} \tag{3.35}$$

This implicit scheme is second order and can also be written:

$$\frac{3}{2}\frac{u^{n+1} - u^n}{\Delta t} - \frac{1}{2}\frac{u^n - u^{n-1}}{\Delta t} = f^{n+1} \tag{3.36}$$

For the system $\frac{du}{dt} = \lambda u$, the scheme is stable if

$$u^{n+1} = \rho u^n = \rho^2 u^{n-1} \Longrightarrow 3\rho^2 - 4\rho + 1 = 2\lambda\Delta t \rho^2 \Longrightarrow \rho = \frac{2 \pm \sqrt{1 + 2\lambda\Delta t}}{3(1 - 2\lambda\Delta t)} \tag{3.37}$$

Fourier document

TODO.

Euler equations for compressible flows

The Euler equations are the following

$$
\begin{aligned}
\frac{\partial \rho}{\partial t} + \sum_j \frac{\partial}{\partial x_j}(\rho u_j) &= 0 \\
\frac{\partial}{\partial t}(\rho u_i) + \sum_j \frac{\partial}{\partial x_j}(\rho u_i u_j) &= -\frac{\partial p}{\partial x_i} + \rho g_i \\
\frac{\partial}{\partial t}(\rho U_0) + \sum_j \frac{\partial}{\partial x_j}(\rho U_0 u_j) &= -\sum_j \frac{\partial}{\partial x_j}(\rho u_j)
\end{aligned} \tag{5.1}
$$

The first one is the conservation of mass and the third one is the conservation of energy. The variables are:

Variable	Name	Variable	Name
ρ	Density	u_i	Velocity in direction i (x, y, z)
x_i	x, y, z	p	Pressure
$g = -\nabla W$	Gravity	$W = gz$	Potential energy
$U_0 = U + K + W$	Total internal energy	$H_0 = H + K + W$	Total enthalpy
U	Internal energy	$H = U + p/\rho$	Enthalpy
$K = \sum_i \frac{u_i^2}{2}$	Kinetic energy		

For an ideal gas, $p = \rho RT$. Then, $dU = C_V(T)dT$ and $dH = C_P(T)dT$ wwith $C_P(T) - C_V(T) = R$. If the gas is perfect, then C_P and C_V are constant and $\gamma := \frac{C_P}{C_V}$. This yields $p = (\gamma - 1)\rho U$. Let us define the following vectors:

$$
\begin{gathered}
\boldsymbol{s} = \begin{bmatrix} s_1 \\ s_2 \\ s_3 \\ s_4 \\ s_5 \end{bmatrix} = \begin{bmatrix} \rho \\ \rho u \\ \rho v \\ \rho w \\ \rho U_0 \end{bmatrix} \qquad \boldsymbol{r}(\boldsymbol{s}) = \begin{bmatrix} 0 \\ 0 \\ 0 \\ -\rho g \\ 0 \end{bmatrix} \\
\boldsymbol{F}_x(\boldsymbol{s}) = \begin{bmatrix} \rho u \\ \rho u^2 + p \\ \rho v u \\ \rho w u \\ \rho H_0 u \end{bmatrix} \qquad \boldsymbol{F}_y(\boldsymbol{s}) = \begin{bmatrix} \rho v \\ \rho u v \\ \rho v^2 + p \\ \rho w v \\ \rho H_0 v \end{bmatrix} \qquad \boldsymbol{F}_z(\boldsymbol{s}) = \begin{bmatrix} \rho w \\ \rho u w \\ \rho v w \\ \rho w^2 + p \\ \rho H_0 w \end{bmatrix}
\end{gathered} \tag{5.2}
$$

Then, the system of equations is

$$\frac{\partial}{\partial t}\boldsymbol{s} + \frac{\partial}{\partial x}\boldsymbol{F}_x(\boldsymbol{s}) + \frac{\partial}{\partial y}\boldsymbol{F}_y(\boldsymbol{s}) + \frac{\partial}{\partial z}\boldsymbol{F}_z(\boldsymbol{s}) = \boldsymbol{r}(\boldsymbol{s}) \in \mathbb{R}^5 \tag{5.3}$$

In 2D, the equation middle equation contains only two equations and all terms in z are 0. In 1D, it is further reduced in the same manner.
In 2D, by the chain rule, we can express this:

$$\frac{\partial \boldsymbol{s}}{\partial t} + \frac{\partial \boldsymbol{F}_x(\boldsymbol{s})}{\partial \boldsymbol{s}}\frac{\partial \boldsymbol{s}}{\partial x} + \frac{\partial \boldsymbol{F}_y(\boldsymbol{s})}{\partial \boldsymbol{s}}\frac{\partial \boldsymbol{s}}{\partial y} = 0 \tag{5.4}$$

where we define the Jacobian matrices $A_x(\boldsymbol{s}) := \frac{\partial \boldsymbol{F}_x(\boldsymbol{s})}{\partial \boldsymbol{s}}$ and $A_y(\boldsymbol{s}) := \frac{\partial \boldsymbol{F}_y(\boldsymbol{s})}{\partial \boldsymbol{s}}$.
In 1D, the Jacobian matrix $A(\boldsymbol{s})$ has three real and distinct eigenvalues:

$$\left.\frac{dx}{dt}\right|_1 = \mu_1 = u - c \qquad \left.\frac{dx}{dt}\right|_2 = \mu_2 = u \qquad \left.\frac{dx}{dt}\right|_3 = \mu_3 = u + c \tag{5.5}$$

where c is the speed of sound defined by $c^2 = \left.\frac{\partial p}{\partial \rho}\right|_S = \gamma\frac{p}{\rho} = \gamma RT$ (last expressions for perfect gas). By the eigenvalues, the system is thus hyperbolic.

- In subsonic flows, $u < c$ and two characteristics go the right while one goes to the left. Therefore, one of the four boundary conditions is applied at the outflow while the other 3 are applied at the inflow.
- In supersonic flows, $u > c$ and all characteristics go in the same direction. Then, all conditions are applied at the inflow.

5.1 Isentropic 1D flow

In isentropic 1D flows, we have the following relation between pressure and density: $dp = c^2 d\rho$ from the relation $c^2 = \left.\frac{\partial p}{\partial \rho}\right|_S$. The equations become

$$\begin{aligned} \frac{\partial \rho}{\partial t} + \frac{\partial}{\partial x}(\rho u) &= 0 \\ \frac{\partial}{\partial t}(\rho u) + \frac{\partial}{\partial x}(\rho u^2) &= -\frac{\partial p}{\partial x} = -c^2\frac{\partial \rho}{\partial x} \end{aligned} \tag{5.6}$$

This gives a first order ODE system for ρ and u on which we can use the method of characteristics. Those characteristics are $\frac{dx}{dt} = u - c$ and $\frac{dx}{dt} = u + c$, along which the quantities $\frac{(\gamma-1)}{2}u - c$ and $\frac{(\gamma-1)}{2}u + c$ are conserved respectively. They are called the Riemann invariants.

Using the Jacobian matrices, the equations become

$$\frac{\partial}{\partial t}\begin{pmatrix}\rho \\ \rho u\end{pmatrix} + \begin{pmatrix}0 & 1 \\ c^2 - u^2 & 2u\end{pmatrix}\frac{\partial}{\partial x}\begin{pmatrix}\rho \\ \rho u\end{pmatrix} = 0 \tag{5.7}$$

5.1.1 Boundary conditions for 1D flows

For a subsonic inflow, we can impose the enthalpy and entropy at the inflow:

$$\begin{cases} H_{0i} = \left(H + \frac{u^2}{2}\right)_i = \left(H + \frac{u^2}{2}\right)_\infty = H_{0\infty} \\ S_{0i} = S_{0\infty} \Longleftrightarrow \frac{p_i}{p_\infty} = \left(\frac{\rho_i}{\rho_\infty}\right)^\gamma \end{cases} \tag{5.8}$$

The subscript i is for inflow and ∞ is for upstream. From those conditions, the density ρ is extrapolated from the interior of the computational domain: $\rho_1 = \rho_2$.

For the corresponding subsonic outflow, we impose a single condition: for example the pressure $p_N = p_d$, for N the index of the last point of the domain, and d for downstream. Then ρu and ρU_0 are extrapolated from the previous value of the computational domain. Any of the three values can be imposed depending on the situation, but that changes of course the solution.

For a supersonic flow, everything is imposed at the inlet and everything is extrapolated at the outlet.

Hyperbolic systems

The Euler equations are a hyperbolic system. Its conservative form in 1D is on the left and the non conservative on the right:

$$\frac{\partial \boldsymbol{s}}{\partial t} + \frac{\partial \boldsymbol{F}(\boldsymbol{s})}{\partial x} = 0 \qquad\qquad \frac{\partial \boldsymbol{s}}{\partial t} + A(\boldsymbol{s})\frac{\partial \boldsymbol{s}}{\partial x} = 0 \tag{6.1}$$

For the system to be hyperbolic, the matrix $A(\boldsymbol{s})$ must have real and distinct eigenvalues. The simplest case of hyperbolic system is the convection equation:

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = 0 \tag{6.2}$$

6.1 Explicit Euler scheme

The explicit Euler scheme both in time and space yields

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = -c\frac{u_{i+1}^n - u_{i-1}^n}{2h} \Longleftrightarrow u_i^{n+1} = u_i^n - \frac{c\Delta t}{2h}\left(u_{i+1}^n - u_{i-1}^n\right) \tag{6.3}$$

This approximation is actually the following ODE:

$$\begin{aligned} \left.\frac{\partial u}{\partial t}\right|_i^n + \frac{\Delta t}{2}\left.\frac{\partial^2 u}{\partial t^2}\right|_i^n + \cdots = -c\left(\left.\frac{\partial u}{\partial x}\right|_i^n + \frac{h^2}{6}\left.\frac{\partial^3 u}{\partial x^3}\right|_i^n + \ldots\right) \\ \Longleftrightarrow \frac{\partial u}{\partial t} = -c\frac{\partial u}{\partial x} - \frac{\Delta t}{2}\frac{\partial^2 u}{\partial t^2} - \frac{ch^2}{6}\frac{\partial^3 u}{\partial x^3} + \ldots \end{aligned} \tag{6.4}$$

From this, we can express the second derivative $\frac{\partial^2 u}{\partial x^2}$ by derivating the equation once. This finally yields

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = \frac{-c\Delta t^2}{2}\frac{\partial^2 u}{\partial x^2} + \ldots \tag{6.5}$$

The Euler explicit scheme produces a diffusion term with negative coefficient. This is unstable and thus the scheme does not work.

6.2 Lax scheme

The Lax scheme stabilizes the Euler explicit scheme by replacing u_i^n by $\frac{1}{2}\left(u_{i+1}^n + u_{i-1}^n\right)$. It yields the following:

$$u_i^{n+1} = \frac{1}{2}\underbrace{\left(u_{i+1}^n + u_{i-1}^n\right)}_{=u_i^n + \frac{1}{2}\left(u_{i+1}^n - 2u_i^n + u_{i-1}^n\right)} - \frac{c\Delta t}{2h}\left(u_{i+1}^n - u_{i-1}^n\right) \tag{6.6}$$

where the underbraced term comes from the Taylor series of u_{i+1}^n and u_{i-1}^n around u_i^n. The modified equation is

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = -\frac{c^2\Delta t}{2}\frac{\partial^2 u}{\partial x^2} + \frac{h^2}{2\Delta t}\frac{\partial^2 u}{\partial x^2} + \dots = \frac{h^2}{2\Delta t}\left(1 - \left(\frac{c\Delta t}{h}\right)^2\right)\frac{\partial^2 u}{\partial x^2} + \dots \tag{6.7}$$

If $\beta = \frac{|c|\Delta t}{h} < 1$, then the diffusion coefficient is positive and the scheme is stable. The right hand side of the equation is some numerical diffusion induced by the method. The RHS can be re-expressed as follows:

$$\frac{|c|h}{2}\frac{(1-\beta^2)}{\beta}\frac{\partial^2 u}{\partial x^2} + \dots \tag{6.8}$$

At $\beta < 1$ fixed, the scheme is convergent when h tends to 0. However, at h fixed, the diffusion term explodes when Δt becomes small. This behaviour is very bad.

6.3 Explicit Euler scheme and upwinding

Upwinding consists in using decentered finite differences to improve stability. For a positive velocity ($c > 0$), the explicit Euler scheme with upwinding becomes

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = -c\frac{u_i^n - u_{i-1}^n}{h} \tag{6.9}$$

The u_{i-1}^n is the upwinding. The modified equation is then

$$\begin{aligned}\frac{\partial u}{\partial t} + \frac{\Delta t}{2}\frac{\partial^2 u}{\partial t^2} + \dots &= -c\left(\frac{\partial u}{\partial x} - \frac{h}{2}\frac{\partial^2 u}{\partial x^2} + \dots\right)\\ \frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} &= \frac{ch}{2}\left(1 - \frac{c\Delta t}{h}\right)\frac{\partial^2 u}{\partial x^2} + \dots\end{aligned} \tag{6.10}$$

In the case of a negative velocity ($c < 0$), the upwinding yields

$$\frac{u_i^{n+1} - u_i^n}{\Delta t} = -c\frac{u_{i+1}^n - u_i^n}{h} \tag{6.11}$$

And the modified equation is

$$\frac{\partial u}{\partial t} + c\frac{\partial u}{\partial x} = \frac{|c|h}{2}(1-\beta)\frac{\partial^2 u}{\partial x^2} + \dots \tag{6.12}$$

This is exactly the same as for a positive velocity. Then, if $\beta < 1$, the diffusion coefficient is positive and the method is stable. At h fixed, the diffusion still increases when Δt becomes small, but much less than the Lax scheme.

6.4 Lax-Wendroff scheme

The Lax-Wendroff scheme is obtained by using the Taylor series up to second order terms: from Equation (6.1),

$$\frac{\partial^2 s}{\partial t^2} = -\frac{\partial}{\partial t}\frac{\partial \boldsymbol{F}(\boldsymbol{s})}{\partial x} = -\frac{\partial}{\partial x}\frac{\partial \boldsymbol{F}(\boldsymbol{s})}{\partial t} = -\frac{\partial}{\partial x}\left(\frac{\partial \boldsymbol{F}(\boldsymbol{s})}{\partial \boldsymbol{s}}\frac{\partial \boldsymbol{s}}{\partial t}\right) = \frac{\partial}{\partial x}\left(A(\boldsymbol{s})\frac{\partial \boldsymbol{F}(\boldsymbol{s})}{\partial x}\right) \tag{6.13}$$

The Lax-Wendroff scheme uses

$$s_i^{n+1} = s_i^n + \Delta t \left.\frac{\partial s}{\partial t}\right|_i^n + \frac{(\Delta t)^2}{2} \left.\frac{\partial^2 s}{\partial t^2}\right|_i^n + \dots \tag{6.14}$$

Then the scheme is

$$s_i^{n+1} = s_i^n - \Delta t \frac{F_{i+1}^n - F_{i-1}^n}{2h} + \frac{(\Delta t)^2}{2} \frac{A_{i+1/2}^n \left(F_{i+1}^n - F_i^n\right) - A_{i-1/2}^n \left(F_i^n - F_{i-1}^n\right)}{h^2} \tag{6.15}$$

where either $A_{i+1/2} = \frac{A(s_i^n)+A(s_{i+1}^n)}{2}$ or $A_{i+1/2} = A\left(\frac{s_i^n+s_{i+1}^n}{2}\right)$. Because the scheme uses Taylor series up to order 2, there is no numerical diffusion. Furthermore, the truncation error is of order $\mathcal{O}(h^2, (\Delta t)^2)$. The scheme is stable if $\beta = \frac{|\mu_k|\Delta t}{h} < 1$, where μ_k are the eigenvalues of $A(s)$.

For simple convection in 1D, the Lax Wendroff scheme yields

$$u_n^{n+1} = u_i^n - c\Delta t \frac{u_{i+1}^n - u_{i-1}^n}{2h} + \frac{1}{2} c^2 (\Delta t)^2 \frac{u_{i+1}^n - 2u_i^n + u_{i-1}^n}{h^2} \tag{6.16}$$

The Lax-Wendroff scheme is not very convenient to use as we need to evaluate $A(s)$. That is not a problem if we have the analytical formulation, but it is rarely the case. There exist some methods using two steps that are equivalent to Lax-Wendroff that are easier to use.

6.5 Richtmeyer scheme

The Richtmeyer scheme uses a Lax-scheme predictor and a leapfrog-scheme corrector:

$$\begin{cases} s_{i+1/2}^* = \frac{1}{2}(s_{i+1} + s_i^n) - \frac{\Delta t}{2} \frac{F_{i+1}^n - F_i^n}{h} \\ s_i^{n+1} = s_i^n - \Delta t \frac{F_{i+1/2}^* - F_{i-1/2}^*}{h} \end{cases} \tag{6.17}$$

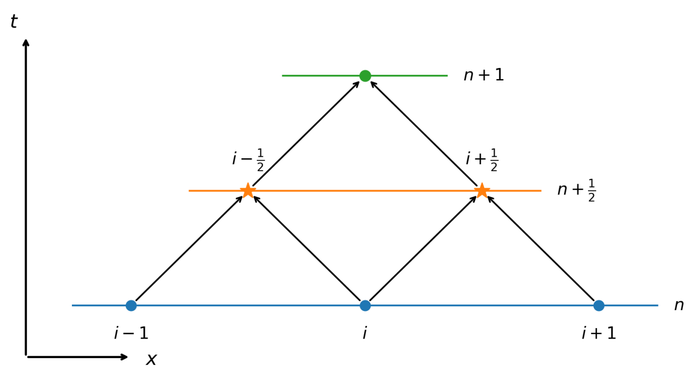


Figure 6.1: Illustration of the Richtmeyer scheme.

6.6 McCormack scheme

The McCormack scheme uses forward and backward decentered finite differences as predictor and corrector. The equations are the following:

$$\begin{cases} \boldsymbol{s}_i^* = \boldsymbol{s}_i^n - \Delta t \frac{\boldsymbol{F}_{i+1}^n - \boldsymbol{F}_i^n}{h} \\ \boldsymbol{s}_i^{n+1} = \frac{1}{2}\boldsymbol{s}_i^* + \frac{1}{2}\left(\boldsymbol{s}_i^n - \Delta t \frac{\boldsymbol{F}_i^* - \boldsymbol{F}_{i-1}^*}{h}\right) = \boldsymbol{s}_i^n - \frac{\Delta t}{2}\left(\frac{\boldsymbol{F}_{i+1}^n - \boldsymbol{F}_i^n}{h} + \frac{\boldsymbol{F}_i^* - \boldsymbol{F}_{i-1}^*}{h}\right) \end{cases} \tag{6.18}$$

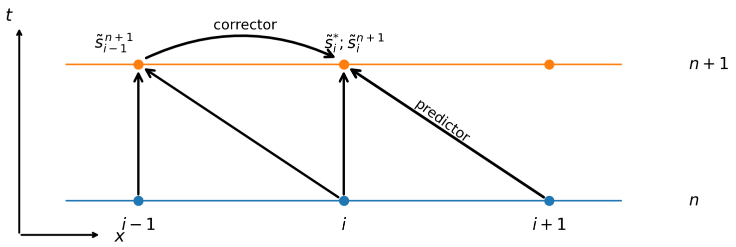


Figure 6.2: Illustration of the McCormack scheme.

6.7 Complex physical domains

Consider a complex physical domain with a structured grid, and consider a rectangular computational domain such that $\xi_i = i$, $\eta_j = j$ and $\zeta_k = k$. We define the jacobian matrix of the transformation between the physical domain and the computational domain as

$$J = \begin{pmatrix} \frac{\partial x}{\partial \xi} & \frac{\partial x}{\partial \eta} & \frac{\partial x}{\partial \zeta} \\ \frac{\partial y}{\partial \xi} & \frac{\partial y}{\partial \eta} & \frac{\partial y}{\partial \zeta} \\ \frac{\partial z}{\partial \xi} & \frac{\partial z}{\partial \eta} & \frac{\partial z}{\partial \zeta} \end{pmatrix} \qquad J^{-1} = \begin{pmatrix} \frac{\partial \xi}{\partial x} & \frac{\partial \xi}{\partial y} & \frac{\partial \xi}{\partial z} \\ \frac{\partial \eta}{\partial x} & \frac{\partial \eta}{\partial y} & \frac{\partial \eta}{\partial z} \\ \frac{\partial \zeta}{\partial x} & \frac{\partial \zeta}{\partial y} & \frac{\partial \zeta}{\partial z} \end{pmatrix} \tag{6.19}$$

If we have a hyperbolic system in the physical domain in conservative form Equation (6.1), in the computational domain it becomes

$$\begin{aligned} \frac{\partial}{\partial t}(D\boldsymbol{s}) + \frac{\partial}{\partial \xi}\left(D\left(\frac{\partial \xi}{\partial x}\boldsymbol{F}_x(\boldsymbol{s}) + \frac{\partial \xi}{\partial y}\boldsymbol{F}_y(\boldsymbol{s}) + \frac{\partial \xi}{\partial z}\boldsymbol{F}_z(\boldsymbol{s}) + \frac{\partial \xi}{\partial t}\boldsymbol{s}\right)\right) & \\ + \frac{\partial}{\partial \eta}\left(D\left(\frac{\partial \eta}{\partial x}\boldsymbol{F}_x(\boldsymbol{s}) + \frac{\partial \eta}{\partial y}\boldsymbol{F}_y(\boldsymbol{s}) + \frac{\partial \eta}{\partial z}\boldsymbol{F}_z(\boldsymbol{s}) + \frac{\partial \eta}{\partial t}\boldsymbol{s}\right)\right) & \\ + \frac{\partial}{\partial \xi}\left(D\left(\frac{\partial \zeta}{\partial x}\boldsymbol{F}_x(\boldsymbol{s}) + \frac{\partial \zeta}{\partial y}\boldsymbol{F}_y(\boldsymbol{s}) + \frac{\partial \zeta}{\partial z}\boldsymbol{F}_z(\boldsymbol{s}) + \frac{\partial \zeta}{\partial t}\boldsymbol{s}\right)\right) &= 0 \end{aligned} \tag{6.20}$$

6.8 Burgers equation

The Burgers equation is the simplest example of a nonlinear hyperbolic equation

$$\frac{\partial u}{\partial t} + u\frac{\partial u}{\partial x} = 0 \iff \frac{\partial u}{\partial t} + \frac{\partial}{\partial x}\left(\frac{u^2}{2}\right) = 0 \tag{6.21}$$

The first equation is the advective form while the second is the conservative form. We define the momentum and the energy of the solution:

$$\begin{cases} I := \int_a^b u dx \\ E := \int_a^b \frac{u^2}{2} dx \end{cases} \tag{6.22}$$

where a, b are the bounds of the domain. The energy is conserved as long as u is continuous and decreases as long as it becomes discontinuous.

By the method of characteristics, they are $\frac{dx}{dt} = u(x,t)$ and u is conserved along each characteristic as the equation is homogeneous. Then, the characteristics are straight lines of changing slope.

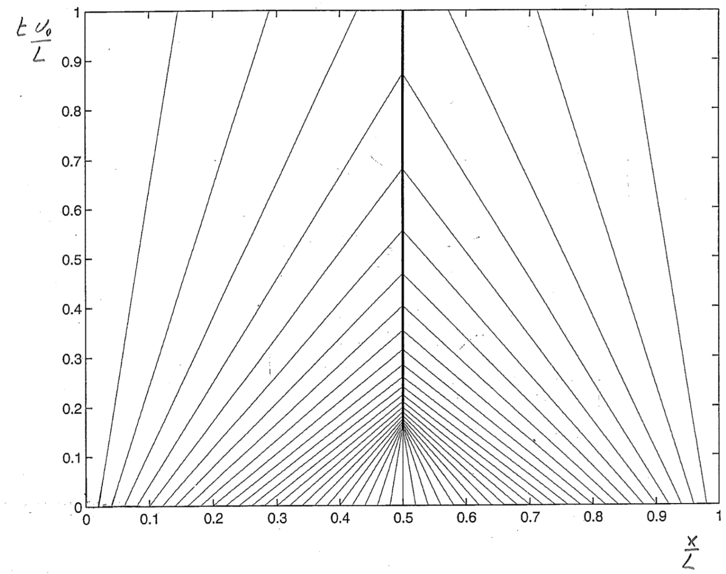
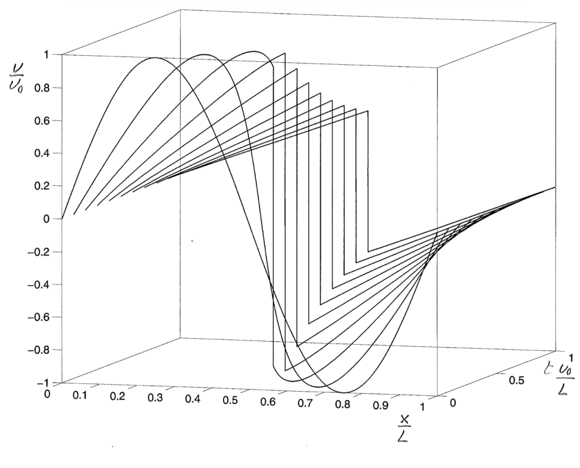
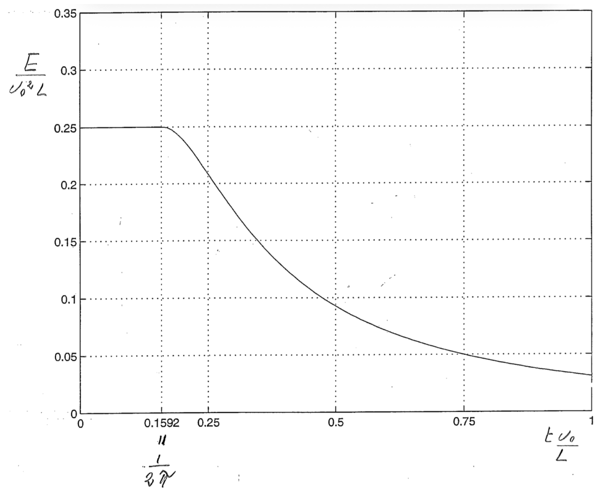


Figure 6.3: Characteristics of the Burgers equation



(a) Solution of the Burgers equation with sine initial condition. A discontinuity appears after a certain time.



(b) The energy is conserved in the Burgers equation until the discontinuity appears.

The shock at the discontinuity conserves the momentum but dissipates energy and creates entropy.

6.8.1 Steady and unsteady discontinuities

The general conservative form is

$$\frac{\partial u}{\partial t} + \frac{\partial}{\partial x}(f(u)) = 0 \tag{6.23}$$

Consider a discontinuity with $u(x_g) = u_g$ on the left and $u(x_d) = u_d$. Then,

$$0 = \int_{x_g}^{x_d} u dx \tag{6.24}$$

Then,

$$\frac{d}{dt}\left(\int_{x_g}^{x_d} u dx\right) = 0 \Longrightarrow \underbrace{\int_{x_g}^{x_d} \frac{\partial u}{\partial t} dx}_{=-\int_{x_g}^{x_d} \frac{\partial}{\partial x}(f(u))dx = -(f(u_d)-f(u_g))} + u_d \underbrace{\frac{dx_d}{dt}}_{\frac{dx_c}{dt}} - u_g \underbrace{\frac{dx_g}{dt}}_{\frac{dx_c}{dt}} \tag{6.25}$$

Then,

$$(u_g - u_d)\frac{dx_c}{dt} = f(u_g) - f(u_d) \tag{6.26}$$

where $\frac{dx_c}{dt}$ is the velocity of the discontinuity. This equation is the **jump condition**. For Burgers, it is

$$\frac{dx_c}{dt} = \frac{u_g + u_d}{2} \tag{6.27}$$

6.8.2 Simulation using finite differences

We can do three different schemes:

$$\begin{aligned} \frac{du_i}{dt} &= -u_i \frac{u_{i+1} - u_{i-1}}{2h} \\ \frac{du_i}{dt} &= -\frac{1}{2}\frac{u_{i+1}^2 - u_{i-1}^2}{2h} \\ \frac{du_i}{dt} &= \frac{1}{3}(a) + \frac{2}{3}(b) \end{aligned} \tag{6.28}$$

By computing $\frac{dI}{dt}$ and $\frac{dE}{dt}$, we can show that all schemes conserve I, but only the third conserves energy (even though it should not because of the discontinuity). To make this discontinuity appear, we need some numerical diffusion in the region of the discontinuity, for example a nonlinear term that is negligible away from the discontinuity, e.g.

$$\frac{C}{h}\left(|u_{i+1} - u_i|(u_{i+1} - u_i) - |u_i - u_{i-1}|(u_i - u_{i-1})\right) \tag{6.29}$$

for some dimensionless coefficient $C > 0$ calibrated by experimentations. The addition of this term does not modify the conservation of momentum.

6.8.3 Explicit Euler scheme with upwinding

We need to check the cases of the sign of u_i^n just like for the convection, as the velocity corresponds to u here. If $u_i^n > 0$, then the explicit Euler scheme with upwinding yields

$$\begin{aligned} \frac{u_i^{n+1} - u_i^n}{\Delta t} &= -u_i \frac{u_i^n (u_i^n - u_{i-1}^n)}{h} \\ &= -(u_i^n + u_{i-1}^n) \frac{u_i^n - u_{i-1}^n}{2h} \end{aligned} \tag{6.30}$$

If $u_i^n < 0$, then

$$\begin{aligned} \frac{u_i^{n+1} - u_i^n}{\Delta t} &= -u_i^n \frac{u_{i+1}^n - u_i^n}{h} \\ &= -(u_{i+1}^n + u_i^n) \frac{u_{i+1}^n - u_i^n}{h} \end{aligned} \tag{6.31}$$

This scheme avoid the oscillations before and after the discontinuity that appeared in the previous scheme.

Numerical methods for incompressible flows

Let us consider single phase flows: $\rho = \rho_0$. Then, the Navier-Stokes equations are

$$\begin{aligned} \nabla \cdot \boldsymbol{u} &= 0 \\ \rho_0 \frac{D\boldsymbol{u}}{Dt} &= -\nabla p + \rho_0 \boldsymbol{g} + \nabla \cdot \tau \\ \rho_0 c \frac{DT}{Dt} &= \Phi - \nabla \cdot \boldsymbol{q} \end{aligned} \tag{7.1}$$

where the first equation comes from continuity and is a constraint, while the others are the conservation of momentum and energy of the system. $\tau = 2\mu d = 2\mu \frac{\nabla \boldsymbol{u} + \nabla \boldsymbol{u}^T}{2}$ is the shear stress tensor, $\boldsymbol{q} = -k\nabla T$ the heat flux, c the specific heat and $\Phi = \tau \cdot d > 0$ the viscous dissipation. As $\boldsymbol{g} = -g\hat{\boldsymbol{e}}_z$, we obtain

$$\begin{aligned} \rho_0 \frac{D\boldsymbol{u}}{Dt} &= -\nabla \left(p + \rho_0 W\right) + \nabla \cdot (2\mu d) \\ \rho_0 c \frac{DT}{Dt} &= 2\mu d \cdot d + \nabla \cdot (k \nabla T) \end{aligned} \tag{7.2}$$

In general, the parameters depend on the temperature. If we consider that μ is not, then we can solve the fluid mechanics without having to solve the temperature. In that case, we have

$$\begin{aligned} \nabla \cdot \boldsymbol{u} &= 0 \\ \rho_0 \frac{D\boldsymbol{u}}{Dt} &= -\nabla \left(p + \rho_0 W\right) + \mu \nabla^2 \boldsymbol{u} \end{aligned} \tag{7.3}$$

We define the kinematic pressure $P := \frac{p + \rho_0 W}{\rho_0}$ and the kinematic viscosity $\nu = \mu / \rho$. This yields

$$\boxed{\begin{aligned} \nabla \cdot \boldsymbol{u} &= 0 \\ \frac{D\boldsymbol{u}}{Dt} &= -\nabla P + \nu \nabla^2 \boldsymbol{u} \end{aligned}} \tag{7.4}$$

With the advective form

$$\frac{D\boldsymbol{u}}{Dt} = \frac{\partial \boldsymbol{u}}{\partial t} + (\nabla \boldsymbol{u}) \cdot \boldsymbol{u} = \frac{\partial \boldsymbol{u}}{\partial t} + A \tag{7.5}$$

and the divergence form

$$\frac{D\boldsymbol{u}}{Dt} = \frac{\partial \boldsymbol{u}}{\partial t} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = \frac{\partial \boldsymbol{u}}{\partial t} + A \tag{7.6}$$

$\rightarrow$ Note: both formulations are physically equivalent, but not numerically.

7.1 2D flows

For 2D flows, the Navier-Stokes equations from Equation (7.4)

$$\begin{aligned}
\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y} &= 0 \\
\frac{\partial u}{\partial t} + a + \frac{\partial P}{\partial x} &= \nu \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) \\
\frac{\partial v}{\partial t} + b + \frac{\partial P}{\partial y} &= \nu \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right)
\end{aligned} \tag{7.7}$$

where a and b are

	Advective form	Divergence form
a	$u\frac{\partial u}{\partial x} + v\frac{\partial u}{\partial y}$	$\frac{\partial}{\partial x}(uu) + \frac{\partial}{\partial y}(uv)$
b	$u\frac{\partial v}{\partial x} + v\frac{\partial v}{\partial y}$	$\frac{\partial}{\partial x}(vu) + \frac{\partial}{\partial y}(vv)$

→ Note: For any vector a, $\nabla \times (\nabla \times a) = -\nabla \cdot (\nabla a) + \nabla(\nabla \cdot a)$. Here, because of incompressibility, $\nabla \times (\nabla \times u) = -\nabla \cdot (\nabla u) + \nabla \underbrace{(\nabla \cdot u)}_{=0}$. This gives $\nabla^2 u = -\nabla \times \omega$, for $\omega := \nabla \times u$ the vorticity.

→ Note: the initial and boundary conditions must be consistent with the constraint $\nabla \cdot \boldsymbol{u} = 0$.

Because of the incompressibility constraint, the problem is elliptic: all points are connected to all other points.

7.1.1 Iterative methods for the case of steady flows

We wish to obtain the solution of the following system:

$$\begin{aligned}
\boldsymbol{A} + \nabla P &= \nu \nabla^2 \boldsymbol{u} \\
\nabla \cdot \boldsymbol{u} &= 0
\end{aligned} \tag{7.8}$$

The Chorin method consists in adding a pseudo-time τ and iterate until convergence of the system:

$$\begin{aligned}
\frac{\partial \boldsymbol{u}}{\partial \tau} + \boldsymbol{A} + \nabla P &= \nu \nabla^2 \boldsymbol{u} \\
\frac{\partial P}{\partial \tau} + c_0^2 \nabla \cdot \boldsymbol{u} &= 0
\end{aligned} \tag{7.9}$$

The parameter c_0^2 has dimensions $[L^2/T^2]$. It is a velocity and its choice should be made in relation to the characteristic velocity of the flow.

7.1.2 Discretization

The discretization is done using a staggered "marker and cell" (MAC) mesh.

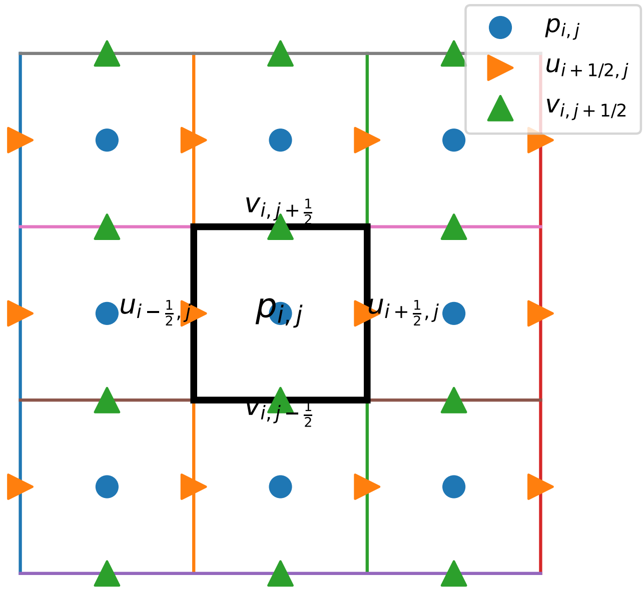


Figure 7.1: Discretization of the field to solve steady-state Navier-Stokes in 2D.

The Euler explicit iteration is

$$
\begin{aligned}
&\frac{u^{n+1}_{i+1/2,j} - u^n_{i+1/2,j}}{\Delta t} + a^n_{i+1/2,j} + \frac{P^n_{i+1,j} - P^n_{i,j}}{\Delta x} \\
&\qquad = \nu \left(\frac{u^n_{i+3/2,j} - 2u^n_{i+1/2,j} + u^n_{i-1/2,j}}{(\Delta x)^2} + \frac{u^n_{i+1/2,j+1} - 2u^n_{i+1/2,j} + u^n_{i+1/2,j-1}}{(\Delta y)^2} \right) \\
&\frac{v^{n+1}_{i,j+1/2} - v^n_{i,j+1/2}}{\Delta t} + b^n_{i,j+1/2} + \frac{P^n_{i,j+1} - P^n_{i,j}}{\Delta y} \\
&\qquad = \nu \left(\frac{v^n_{i+1,j+1/2} - 2v^n_{i,j+1/2} + v^n_{i-1,j+1/2}}{(\Delta x)^2} + \frac{v^n_{i,j+3/2} - 2v^n_{i,j+1/2} + v^n_{i,j-1/2}}{(\Delta y)^2} \right)
\end{aligned}
\tag{7.10}
$$

To impose the initial and boundary conditions on the MAC mesh, we add ghost points on the outside of the physical domain. Furthermore, as P is defined in the middle of the cells, there is no boundary condition on it. We need to impose the ghost value so that the first and second derivative of the velocity is accurate to $\mathcal{O}((\Delta x)^2)$ so that the solution on the rest of the grid can be accurate too. Let us denote v_0 the value of the ghost point. Then, by the Taylor series, we have

v_0	$\left.\frac{\partial v}{\partial x}\right\vert_1 = \frac{v_2 - v_0}{2\Delta x}$	$\left.\frac{\partial^2 v}{\partial x^2}\right\vert_1 = \frac{v_2 - 2v_1 + v_0}{(\Delta x)^2}$
$-(v_1 - 2v_\Gamma)$	$\mathcal{O}(\Delta x)$	Incorrect
$\frac{1}{3}(v_2 - 6v_1 + 8v_\Gamma)$	$\mathcal{O}((\Delta x)^2)$	$\mathcal{O}(\Delta x)$
$-\frac{1}{5}(v_3 - 5v_2 + 15v_1 - 16v_\Gamma)$	$\mathcal{O}((\Delta x)^2)$	$\mathcal{O}((\Delta x)^2)$

The equation for the pressure in steady state incompressible flows is

$$\nabla^2 P = -\nabla \cdot \boldsymbol{A} \tag{7.11}$$

7.1.3 Methods for unsteady flows

The following explicit scheme of order $\mathcal{O}(\Delta t)$

$$\begin{cases} \frac{\boldsymbol{u}^* - \boldsymbol{u}^n}{\Delta t} + \boldsymbol{A} = \nu \nabla^2 \boldsymbol{u}^n \\ \frac{\boldsymbol{u}^{n+1} - \boldsymbol{u}^*}{\Delta t} + \nabla P = 0 \\ \nabla \cdot \boldsymbol{u}^{n+1} = 0 \end{cases} \tag{7.12}$$

yields the following Poisson equation

$$\nabla^2 P = \frac{1}{\Delta t} \nabla \cdot \boldsymbol{u}^* \tag{7.13}$$